



Exploring the shapes of graphs using technology

- 1 A _____ equation is an equation where the highest exponent of the variable is 2. (Tick 1 correct answer)

- ☐ arithmetic
- ☐ cubic
- ☐ geometric
- ☐ linear
- ☒ quadratic

- 2 Select all the quadratic equations. (Tick 2 correct answers)

- ☐ $y = x + 2$
- ☐ $y = 2x$
- ☒ $y = x^2$
- ☒ $y = x^2 + 3x$
- ☐ $y = x^2 + x^3$

- 3 _____ is a special symmetrical curve which can be bowl-shaped or archlike. (Tick 1 correct answer)

- ☐ A cubic curve
- ☐ An exponent
- ☐ A straight-line
- ☒ A parabola
- ☐ A parallel



4 Which of the below equations will form a parabola when plotted? (Tick 3 correct answers)

☒ $y = x^2$

☒ $y = x^2 + 3x$

☐ $x + y = 2$

☒ $y = -x^2$

☐ $y = x^3 + x^2$

5 Match each equation to the description of its shape when graphed. (Write the correct letter in each box)

a	$2x + 3y = 12$
b	$y = x^2 + 3x$
c	$y = x^3 + 2x$

b	Parabola.
c	Cubic curve.
a	Straight graph.

6 Which of these equations will produce parabolas that are 'archlike' or appear to 'open downwards' when graphed? (Tick 1 correct answer)

☐ $y = x^2 - 5$

☐ $y = x^2 - 5x$

☐ $y = x^2 + 5x$

☒ $y = 5 - x^2$

☐ $y = 5 + x^2$

